

Exam Date & Time: 13-Mar-2026 (11:30 AM - 12:30 PM)



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Mid - Term Theory Examination March 2026
Fourth Semester of B. Tech. (EC)

DATA COMMUNICATION AND NETWORKING [ECUC257]

Marks: 20

Duration: 60 mins.

SECTION - I

Answer all the questions.

1		
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An organization has been assigned the IP address 192.168.10.0/24. The organization requires 4 subnets. Create the four subnets with appropriate subnet mask.

(2)

Course Outcome: "CO3" / Cognitive Level: "Create"

2		
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A packet passes through 4 routers. Find the total delay.

Given:

Transmission Delay = 2 ms

Propagation Delay = 5 ms

Processing Delay = 1 ms

Queuing Delay = 3 ms

(2)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

3		
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Differentiate between logical addressing and port addressing. Why are both required for process-to-process delivery?

(2)

Course Outcome: "CO4" / Cognitive Level: "Remember"

4		
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What is the difference between unicast, broadcast, and multicast communication?

(2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

5		
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Explain any two key differences between Static Routing and Dynamic Routing.

(2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

SECTION - II

Answer all the questions.

1		
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A company wants centralized control of resources. Should it use client-server or peer-to-peer architecture? Justify your answer.

(2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

2		
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Explain the OSI Reference Model with its seven layers. Compare it briefly with the TCP/IP protocol architecture. Use a neat diagram to illustrate the differences.

(5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

3		
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Which transmission method is suitable for minimal overhead and high-speed data transfer of large blocks of data? Briefly state the reason.

(1)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

4		
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What is the primary purpose of the Sliding Window Protocol in data communication?

(2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

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Event Name		Subject	
CHARUSAT Midterm Exam Even 2025-2026		ECUC257:DATA COMMUNICATION AND NETWORKING	
#	Course Outcome	Description	Marks
1	CO1	Explain the fundamentals of data communication and networking, including communication models, transmission media, and protocol architectures.	8
2	CO3	Analyze the operation of network layer functions, including routing algorithms, congestion control, and IP addressing schemes.	8
3	CO4	Differentiate and compare transport layer protocols and demonstrate basic socket programming concepts.	4
Total			20

* NA - CO Not Selected

Event Name		Subject	
CHARUSAT Midterm Exam Even 2025-2026		ECUC257:DATA COMMUNICATION AND NETWORKING	
#	Level	No of Questions	Marks (%)
1	Remember	1	2(10%)
2	Understand	3	9(45%)
3	Application	0	0(0%)
4	Analysis	3	5(25%)
5	Create	1	2(10%)
6	Evaluate	1	2(10%)
Total		9	20 (100%)